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Effects of farmyard manure application and grassland renovation on net ecosystem carbon balance in a temperate grassland: analysis of 11-year eddy covariance data

Shoji MATSUURA^{a,†}, Akinori MORI^b, Akira MIYATA^{a,c} and Ryusuke HATANO^d^a Institute for Agro-Environmental Sciences, National Agriculture and Food Research Organization, 3-1-3 Kannondai, Tsukuba, Ibaraki 305-8604, Japan^b Institute of Livestock and Grassland Science, National Agriculture and Food Research Organization, 768 Senbonmatsu, Nasushiobara, Tochigi 329-2793, Japan^c Bio-oriented Technology Research Advancement Institution, National Agriculture and Food Research Organization, 8 Higashida, Kawasaki, Kanagawa 210-0005, Japan^d Graduate School of Agriculture, Hokkaido University, Kita 9 Nishi 9 Kita-ku, Sapporo, Hokkaido 060-8589, Japan

Abstract

Grasslands provide a globally essential ecosystem service by storing carbon (C). Management practices affect the net C balance in grasslands positively and negatively. In Japan, farmyard manure (FYM) application and grassland renovation are common and essential grassland management practices; however, the long-term effects of these practices on C balance are not known. We conducted a field experiment in a managed grassland for 11 years (2004–2015) to investigate the effects of FYM application and grassland renovation on net ecosystem C balance (NECB), estimated from eddy covariance measurements. Two experimental plots were used: One received only chemical fertilizer (CF plot), and the other received FYM and supplementary chemical fertilizer (CM plot). Grassland renovation with full inversion tillage was conducted twice during the study period. The C annually removed from the grassland through harvest always exceeded the annual net ecosystem production, resulting in a net C loss (NECB = $-1.32 \text{ Mg C ha}^{-1} \text{ y}^{-1}$) in the CF plot. In contrast, the CM plot gained net C (NECB = $1.79 \text{ Mg C ha}^{-1} \text{ y}^{-1}$) due to FYM application. The difference in ecosystem respiration between the plots suggested that 45% of C in annually applied FYM remained in the soil. The FYM application-mediated C input necessary to avoid net C loss was $3.67 \text{ Mg C ha}^{-1} \text{ y}^{-1}$. Grassland renovation conducted twice showed a mixed but relatively minor effect. Grassland renovation had little impact on annual NECB when the degree of sward/soil disturbance was low. In contrast, renovation with a high degree of sward/soil disturbance led to negative NECB in the year of renovation even in the CM plot. Overall, FYM application contributed to a net C gain, and the effect of grassland renovation on annual NECB was likely limited to the years of renovation.

Key words: Carbon dioxide flux, Carbon stock, Managed grassland, Organic matter application, Plowing

1. Introduction

Grasslands are the largest terrestrial ecosystems, covering 31%–43% of the earth's land area (Adams *et al.*, 1990; White *et al.*, 2000; Asner *et al.*, 2004). Grasslands provide many ecosystem services globally, such as biodiversity protection, climate regulation, and soil conservation, in addition to their widespread use in livestock production through grazing or herbage production (Sala and Paruelo, 1997). Carbon (C) storage is one of the essential ecosystem services provided by grasslands (Gibson, 2009; Zhao *et al.*, 2020). Generally, grasslands have a large amount of belowground biomass (Gibson, 2009; Peichl *et al.*, 2011), leading to a high amount of belowground C input and resulting in a relatively high level of soil organic carbon (SOC) compared with other ecosystems (Lal, 2004; Conant *et al.*, 2017).

For example, according to Lal (2004), global estimates of SOC stocks in temperate grasslands are $141\text{--}236 \text{ Mg C ha}^{-1}$, which are relatively large compared to temperate forests ($96\text{--}147 \text{ Mg C ha}^{-1}$) and croplands ($80\text{--}103 \text{ Mg C ha}^{-1}$). The amount of SOC or net C balance in grasslands is affected, positively and negatively, by management practices (Soussana *et al.*, 2007; Conant *et al.*, 2017), including fertilization (Schuman *et al.*, 2002), grazing (Reeder and Schuman, 2002; Schipper *et al.*, 2010; Chang *et al.*, 2016), irrigation (Kelliher *et al.*, 2012), organic amendment application (Ammann *et al.*, 2007; Franzluebbers and Stuedemann, 2009; Ryals *et al.*, 2014), and grassland renovation (Necpalova *et al.*, 2013; Rutledge *et al.*, 2014; Kayser *et al.*, 2018).

In Japan, managed grasslands occupy 13.5% of all agricultural lands, corresponding to 29.6% of the upland field area (Statistics Department, 2022). Despite comprising a large percentage of the total upland field, managed grasslands have not been adequately assessed for changes in SOC stocks or net C balance. This is partly because data on grassland soil and C dynamics have rarely been collected, and systematic studies have not been conducted. Thus, it is necessary to evaluate C dynamics in managed grasslands, particularly concerning management practices, in areas under temperate climates with relatively high precipitation, such as Japan.

Among grassland management practices, farmyard manure

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[†]Corresponding author: shojim@affrc.go.jp

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(FYM) application is common and essential in Japan because the livestock industry is highly dependent on imported feed, which has caused various environmental problems due to inappropriate handling of livestock excreta. The Japanese government encourages livestock farmers to handle livestock excreta correctly and use it for composting and application to grasslands or other agricultural lands, where it contributes to increase in C stocks (Agricultural Production Bureau, 2018). Previously, we investigated the effects of FYM application on C dynamics in a managed grassland site in Japan based on 3 years (2004–2007) of CO₂ flux data (Matsuura *et al.*, 2014). We found that a grassland field that received FYM and supplementary chemical fertilizer gained net C, whereas a field that received only chemical fertilizer lost net C. Similar findings have been reported on the effects of FYM application on grassland net C balance (Shimizu *et al.*, 2009; Hirata *et al.*, 2013; Limin *et al.*, 2015). However, the effects of long-term FYM application have not been adequately investigated. Grassland renovation is another important form of grassland management in Japan because sustaining a stable sward composition can be difficult, particularly in regions where the annual mean air temperature is >12°C, which causes depression of cool-season grasses due to high summer temperature (Agricultural Production Bureau, 2006). Farmers are advised to renovate grasslands to restore declining productivity based on several regional criteria for soil properties and the degree of weed invasion (Agricultural Production Bureau, 2006). In another study, we collected CO₂ flux measurements around grassland renovation work and found that renovation does not significantly affect immediate (hours to days) net C loss (Matsuura *et al.*, 2017). However, little is known about the longer-term (months to years) effects of renovation on C dynamics.

In the present study, we extended our previous work (Matsuura *et al.*, 2014) with additional eight years (2007–2015) of measurements at the same grassland site. Specifically, the effects of FYM application and grassland renovation with full inversion tillage on the net C balance were investigated. In addition, more detailed characteristics of seasonal C dynamics are discussed herein. Conducting analyses with an 11-year extended dataset enabled us to evaluate C dynamics under more meteorological conditions and assess the effects of a wider range of grassland management practices.

2. Materials and methods

2.1 Site description

The study was conducted at the grassland fields at the National Agriculture and Food Research Organization, Institute of

Livestock and Grassland Science, Nasu Research Station, located in Nasushiobara, Tochigi, Japan (36° 55' N, 139° 56' E, 305 m above sea level; Supplementary material 1). The climate at this location is temperate and humid, and the annual mean temperature and precipitation are 12.2°C and 1632 mm, respectively (30-year climate normals for 1991–2020 at the site), with little snow accumulation, even in midwinter. The site is located at an alluvial fan and the soil's gravel content is relatively high (Kurashima *et al.*, 1993). The soil at the site is affected by volcanic ash and classified as Haplic Brown Lowland soil (Haplic Fluvisols [Eutric] in the World Reference Base for Soil Resources; IUSS Working Group WRB (2006); Table 1). The grassland topography is almost flat, and the grassland is categorized as cultivated pastureland/grassland (Allen *et al.*, 2011), i.e., the grass was grown for cutting, and animals were not grazed. Prior to the present study, the grassland was renovated (plowed and reseeded) with *Dactylis glomerata* L. in 2001 and maintained only with chemical fertilizer with an application rate of ~200 kg N ha⁻¹ y⁻¹.

2.2 Experimental design and field management

The field experiment was conducted from November 2004 to October 2015. Two experimental plots were established on two adjacent grassland fields (Supplementary material 1). One plot received only chemical fertilizer (hereafter known as the “CF plot”, 2.3 ha), whereas the other plot received FYM from dairy cattle with an application rate of ~30 Mg ha⁻¹ y⁻¹ (close to the recommended application rate in managed grasslands in the region) on a fresh weight basis, except in 2004 when the rate was 15 Mg ha⁻¹ y⁻¹ (hereafter known as the “CM plot”, 2.4 ha). The FYM was applied to the soil surface annually from 2004 to 2015 after the final cutting in November, except in 2007 and 2012, when both plots were renovated after the third crop; in 2007 and 2012, the FYM was applied to and incorporated into the soil during renovation work in September.

The application rates of chemical fertilizers in the CF plot were determined based on the recommended levels for the region, targeting a nitrogen (N) application rate of 190–210 kg N ha⁻¹ y⁻¹. For the CM plot, chemical fertilizers were used to compensate for the nutrients that were not obtained by the release from the FYM, which was estimated as follows. The N release rate was estimated using Uchida's model developed in Japan (Shiga *et al.*, 1985), whereas the phosphorus and potassium release rates were estimated based on the handbook of animal waste management and utilization in Hokkaido (Research Team for Animal Waste Project, 2004). Chemical fertilizers were applied in three or four installments for each crop growth, i.e., in early spring (mid-March)

Table 1. Soil properties of the topsoil (around 30 cm) for each plot.

Plot	Depth (cm)	C (%)	N (%)	pH (H ₂ O)	Bulk density (Mg m ⁻³)	Sand (%)	Silt (%)	Clay (%)
CF plot	0–17	3.4	0.24	6.1	1.08	61.7	23.3	15.1
	17–32	3.5	0.24	6.2	0.92	60.0	24.8	15.2
CM plot	0–15	3.6	0.30	5.6	0.83	58.7	21.1	20.2
	15–30	3.6	0.29	5.5	0.96	56.7	23.3	20.0

The analysis was conducted in November 2007. Soil samples were collected from one point in each plot, which differed from the sampling points for the analysis in Table 4.

and after each cutting event except after the final cutting.

The grass was cut and harvested four times per year, except in 2007, 2010, and 2012 when the grass was cut three times per year. In 2010, the timings of the cuttings were delayed. In 2007 and 2012, grassland renovation with full inversion tillage was conducted at both plots after the third crop.

The renovation work conducted in 2012 was a standard one, as practiced in Japan, with major procedures, including the use of a glyphosate-based herbicide spray to kill the existing grass and weeds, application of FYM in the CM plot, plowing to a depth of 200–300 mm using a moldboard plow, application of chemical fertilizer, and seeding followed by soil compacting. Large stones were also removed. All the aforementioned operations were conducted carefully over time. In contrast, the renovation work in 2007 was done quickly and roughly, without the use of herbicide spray and the removal of large stones. The renovation work was done over 4 (CF plot) and 11 (CM plot) days in 2007 (from early- to mid-September) and over 31 days in 2012 (from late-August to late-September). The degree of sward/soil disturbance was, therefore, higher in 2012 than that in 2007. The same amount of chemical fertilizer was applied to each plot during renovation in both years to ensure adequate initial grass growth. For additional details on the renovation of the plots, see the work of Matsuura *et al.* (2017).

2.3 Biometric measurements

All the grass in a 1.3×10 m area was clipped at 5 cm above the ground in 8–30 replicates using a reciprocating grass mower with 1.3 m-wide cutter blades. The clipped biomass was sampled and measured to determine its fresh weight. The subsamples of the fresh samples were oven-dried at 70°C for >48 h and weighed. The FYM samples were collected from the stockpile immediately before or after application on the field. After their fresh weight was measured, the FYM samples were air-dried and weighed. Subsequently, each dried grass and FYM sample was ground to measure total C concentration using a dry combustion method via a C/N analyzer (JM1000CN or MT-700; J-Science Lab, Kyoto, Japan or Yanaco, Tokyo, Japan, respectively). The development of plant canopies was quantified by measuring vegetation height with a measuring rod every 1–2 weeks at ~30 points in each plot.

2.4 Soil measurements

Soil samples from five points in each plot were collected from mineral soil horizons down to 30 cm depth, divided into four layers (0–5, 5–10, 10–20, and 20–30 cm). Samples were collected at the beginning of Y10–11, Y12–13, Y13–14, and Y14–15 (in November, before FYM application). In addition, soil samples were collected in July in Y11–12, just before the grassland renovation was conducted in Y12–13.

One core sample from the middle of each layer of each sampling point was taken to determine the bulk density using a 100-mL stainless-steel cylinder. The core samples were oven-dried at 105°C for >24 h and weighed. Then, coarse fragments (gravel and coarse organic matter) in the core samples that were filtered on a 2-mm sieve were oven-dried at 105°C for >24 h and weighed. Composite samples were taken from each whole layer and passed through 2-mm sieve after air-drying. The subsamples

from the sieved soil samples were ground to measure the total C concentration using a dry combustion method via a C/N analyzer (JM1000CN or MT-700; J-Science Lab or Yanaco, respectively).

SOC stocks per unit area down to 30 cm depth were calculated by summing up the products of the soil bulk density of fine fragments < 2 mm (Matsuura *et al.*, 2021), soil C concentration, and layer thickness. We quantified SOC stocks in equivalent soil mass in each plot to eliminate the effects of soil mass changes (Ellert and Bettany, 1995). The soil mass in Y10–11 was chosen as the reference soil mass for each plot.

2.5 Eddy covariance and meteorological measurements

Fluxes of CO₂, sensible heat, and latent heat were measured using eddy covariance systems installed at each plot (Supplementary material 1). These systems, each consisted of a three-dimensional sonic anemometer (CSAT3; Campbell Scientific, Logan, UT, USA), an open-path infrared CO₂/H₂O gas analyzer (LI-7500; Li-Cor, Lincoln, NE, USA), and a datalogger (CR23X or CR1000; Campbell Scientific). Output signals from the sensors were sampled at a frequency of 10 Hz and recorded on a flash memory card with a custom-built data acquisition device (System Kobo, Watari, Japan) or a CompactFlash module (CFM100; Campbell Scientific). The sensor head of the sonic anemometer and infrared gas analyzer were installed at heights of 2.3 and 2.5 m above the ground in the CF and CM plot, respectively. The predominant wind directions were south and north-west and the minimal fetch lengths for these directions were 68 and 76 m for the CF and CM plot, respectively.

Air temperature and relative humidity were measured using a temperature-humidity sensor (HMP-45A; Vaisala, Helsinki, Finland) equipped with a stainless ventilator (PVC-02-DC; Preed, Tokyo, Japan) at a height of 2.5 m. Incident photosynthetic photon flux density (PPFD) was measured at a height of 1.5 m with a quantum sensor (LI-190SA or ML-020P; Li-Cor or EKO, Tokyo, Japan, respectively). Precipitation data were obtained using a tipping bucket rain gage (Nakaasa Instruments, Tokyo, Japan or Model 52202; R. M. Young, Traverse, MI, USA). These various parameters were measured at either of the plots and sampled every 5 or 10 s via the data loggers (CR23X, 21X, or CR1000; Campbell Scientific), and half-hourly averages or integrated values were recorded.

The 10 Hz eddy covariance data were postprocessed to calculate half-hourly CO₂ flux. The calculation procedures used have been described in detail by Matsuura *et al.* (2014) previously. In brief, the corrections for spike values (Vickers and Mahrt, 1997; Mano *et al.*, 2007), coordinate rotation (Wilczak *et al.*, 2001), frequency response losses (Massman, 2000, 2001), and influence of water vapor on sensible heat flux by applying water vapor correction on sonic virtual temperature (Kaimal and Gaynor, 1991) were conducted, and a density term (Webb *et al.*, 1980) was added. The CO₂ storage was estimated from the difference in mean molar densities between two continuous half-hourly flux values, and the net ecosystem CO₂ exchange (NEE) was calculated as the sum of CO₂ flux and CO₂ storage.

The quality control of the fluxes was performed using quality tests based on meteorological and statistical theories (Mano *et al.*, 2007), basic tests for 10 Hz data (Vickers and Mahrt, 1997),

nonstationarity and integral turbulence tests (Foken and Wichura, 1996), and a half-hourly spike test (Papale *et al.*, 2006). The CO₂ flux under low-turbulence conditions was removed when the friction velocity was $< 0.02 \text{ m s}^{-1}$, which was determined based on a statistical method described by Gu *et al.* (2005). Cumulative flux footprints were calculated using the footprint model of Kljun *et al.* (2004) to ensure that most footprint areas were within the minimal fetch of each plot.

2.6 Partitioning and gap-filling of NEE

Net ecosystem production (NEP) is the difference between gross primary production (GPP) and ecosystem respiration (RE); it has the following relationship with NEE:

$$\begin{aligned} -\text{NEE} &= \text{NEP} \\ &= \text{GPP} - \text{RE} \end{aligned} \quad (1)$$

We separated NEE into GPP and RE using nonlinear empirical models implemented through a flux calculation software (Ueyama *et al.*, 2012), with which gaps in NEE were also filled. Night-time RE is equal to night-time NEE because night-time GPP is equal to zero. Daytime RE was estimated from air temperature via extrapolation using the equation of Lloyd and Taylor (1994), with which night-time gaps in half-hourly NEE were filled. Daytime GPP was estimated from observed NEE and estimated daytime RE using Equation 1. Daytime gaps in half-hourly NEE were filled using a light-photosynthesis curve expressed via a nonrectangular hyperbolic equation (Prioul and Chartier, 1977). Parameters for the Lloyd and Taylor equation and nonrectangular hyperbolic equation were determined each day for a 5-day moving window. The partitioning and gap-filling procedures used have been described in detail by Matsuura *et al.* (2014).

2.7 Estimation of annual and seasonal net carbon balance

When considering annual net C balance in managed ecosystems,

we must also consider non-CO₂ origin C gains and losses, such as those from FYM application and cutting. The net C balance after considering these C gains/losses is defined as net ecosystem C balance (NECB) (Chapin *et al.*, 2006; Smith *et al.*, 2010). For the study site, NECB was determined by the following equation:

$$\text{NECB} = \text{NEP} - F_{\text{H}} + F_{\text{FYM}} \quad (2)$$

where F_{H} is the amount of C removed through harvest and F_{FYM} is the amount of C input through FYM application. Equation 2 was applied to evaluate the annual NECB. In this study, C lost via leaching through the soil to groundwater was not determined and neglected for the calculation of NECB. C loss through erosion was also ignored for the NECB calculation because the grassland topography is primarily flat.

In the present study, an “annual” period was defined as the period between the final (third or fourth) cuttings in consecutive years. For example, the year 2005–2006 (hereafter known as Y05–06) indicates the period from the day of the fourth cutting in 2005 to the day before the fourth cutting in 2006. Although the number of days included in each annual period differs (Table 2), the unit “y” representing “year” is used for these periods when discussing annual values. To evaluate C exchange and net C balance during “annual” periods, each period was divided based on the cutting or fertilization schedule as follows: Autumn–winter (from final cutting to early spring fertilization) and the growing season of the first crop (from early spring fertilization to first cutting), second crop (from first to second cutting), third crop (from second to third cutting), and fourth crop (from third to fourth cutting). Typical cutting and fertilization schedules were as follows: Early spring fertilization in mid-March (around the time when daily mean air temperature increased to $> 5.0^{\circ}\text{C}$), first crop cutting in mid-May, second crop cutting in early- to mid-July, third crop cutting in late-August or early-September, and fourth crop cutting in late-October to early-November. Seasonal C exchange

Table 2. Components of annual net ecosystem carbon balance during the study period ($\text{Mg C ha}^{-1} \text{ y}^{-1}$).

Year †	CF plot						CM plot						
	GPP	RE	NEP	F_{H}	F_{FMY}	NECB ‡	GPP	RE	NEP	F_{H}	F_{FMY}	NECB	$\text{NEP} - F_{\text{H}}$
Y04–05 (377)	21.8	17.6	4.26	4.89	0.00	–0.63	22.1	18.1	3.92	5.16	2.06	0.82	–1.24
Y05–06 (352)	18.9	15.5	3.43	4.13	0.00	–0.70	20.3	16.8	3.50	4.30	4.21	3.40	–0.81
Y06–07 (297)	16.6	13.8	2.80	3.92	0.00	–1.12	16.7	14.6	2.11	3.93	3.96	2.14	–1.82
Y07–08 (438) §	26.3	22.0	4.38	5.86	0.00	–1.48	27.3	24.0	3.29	4.98	6.39	4.70	–1.69
Y08–09 (350)	21.6	17.5	4.11	4.90	0.00	–0.79	22.7	19.4	3.35	5.81	5.31	2.84	–2.47
Y09–10 (343/342 ¶)	23.7	19.6	4.04	5.62	0.00	–1.58	25.5	22.2	3.32	6.67	4.32	0.98	–3.35
Y10–11 (386/387 ¶)	24.3	19.9	4.49	5.82	0.00	–1.34	24.2	21.7	2.48	4.90	3.99	1.57	–2.42
Y11–12 (300)	15.8	13.1	2.75	4.80	0.00	–2.05	17.4	15.3	2.13	4.61	3.77	1.30	–2.48
Y12–13 (434) §	27.4	24.4	2.99	6.24	0.00	–3.25	27.0	25.5	1.49	6.50	3.63	–1.38	–5.00
Y13–14 (363)	18.9	16.4	2.52	3.86	0.00	–1.34	22.1	18.9	3.16	4.26	3.48	2.38	–1.10
Y14–15 (365)	24.2	19.7	4.50	4.73	0.00	–0.23	24.1	21.7	2.48	4.80	3.28	0.97	–2.32
Mean	21.8	18.1	3.66	4.98	0.00	–1.32	22.7	19.8	2.84	5.08	4.04	1.79	–2.24
SD	3.83	3.42	0.77	0.82	0.00	0.82	3.50	3.51	0.74	0.89	1.11	1.59	1.18

GPP, gross primary production; RE, ecosystem respiration; NEP, net ecosystem production; F_{H} , the amount of carbon removed through harvest; F_{FYM} , the amount of carbon input through farmyard manure application; NECB, net ecosystem carbon balance.

† The “annual” period was defined as the period between the final (third or fourth) cutting. Therefore, the number of days, which is indicated in the parenthesis, included in each annual period differs depending on the year.

‡ NECB is equal to $\text{NEP} - F_{\text{H}}$ in CF plot.

§ The years of grassland renovation.

¶ Annual period for CF plot/CM plot; cutting date of the final crop in 2010 differed depending on the plots.

(GPP and RE) and seasonal NECB, defined as the difference between NEP and F_H ($NEP - F_H$), were also evaluated for each divided period. However, to simplify the comparison between the plots, F_{FYM} was not considered when evaluating seasonal NECB.

2.8 Statistical analysis

All statistical analyses were performed using R version 4.2.0 (R Core Team, 2022). Analysis of covariance (ANCOVA) was performed to evaluate the effect of the treatment and meteorological factors (air temperature, precipitation, and PPFD) on annual NECB and its components (i.e., GPP, RE, NEP, and F_H) for the 7-year data, except for the years just before the grassland renovation (Y06–07 and Y11–12) and the years of grassland renovation (Y07–08 and Y12–13), for which annual durations differed substantially from other years. Prior to performing the ANCOVA, slopes of the regression lines between the CF and CM plots were compared for each

meteorological factor on NECB and its components to determine whether there are interaction effects between the treatment and the meteorological factors. An interaction term is included for the ANCOVA if there is a significant interaction effect between the treatment and the meteorological factors. The differences between the plots or among sampling events for the SOC concentration of each layer and SOC stocks of 0–30 cm depth were analyzed with the *t*-test or Tukey-Kramer's honestly significant difference test, respectively.

3. Results

3.1 Meteorological conditions

The seasonal air temperature was similar in all the years (Fig. 1; Supplementary material 2). The daily mean air temperature in midsummer was $>25.0^{\circ}\text{C}$ (daily maximum range: 26.1°C – 28.6°C), but in winter, it fell to $<0.0^{\circ}\text{C}$ (daily minimum range: -4.8°C to -0.2°C). Variation in annual mean air

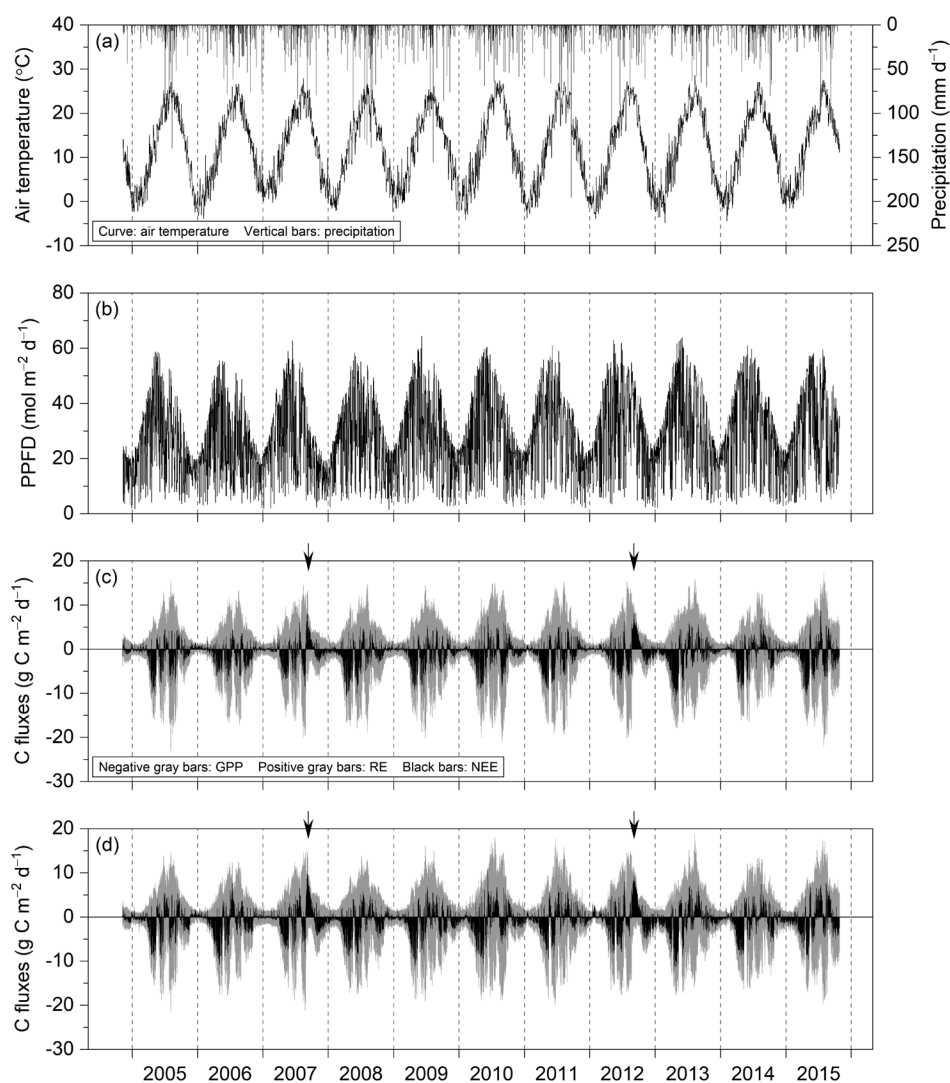


Fig. 1. Seasonal variations in (a) air temperature (curve) and precipitation (vertical bars); (b) incident photosynthetic photon flux density (PPFD); gap-filled daily net ecosystem CO_2 exchange (NEE), gross primary production (GPP), and ecosystem respiration (RE) of the (c) CF plot and (d) CM plot during the study period. NEE is plotted as black bars, whereas GPP and RE are plotted as gray bars (negative values, GPP; positive values, RE). C uptake and C release are denoted as negative and positive values, respectively. Arrows in (c) and (d) indicate the dates of plowing during grassland renovation work.

temperature was relatively small, i.e., 10.1°C – 13.2°C (mean $\pm$ standard deviation (SD): $12.0 \pm 0.78^{\circ}\text{C}$, $n = 11$). Precipitation levels were generally high during summer and early autumn (June–September), particularly during the rainy season (from mid-June to mid-July), but they were lower in winter (December–February). However, seasonal precipitation patterns varied widely among years, and annual precipitation ranged from 1159 to 2375 mm (1636 ± 343 mm). Seasonal changes in PPFD were similar among all the years, with a temporal reduction observed during the rainy season. During the study period, the annual PPFD was 9916 ± 1342 mol m^{-2} .

3.2 Biomass production and applied FYM

Annual F_{H} was 4.98 ± 0.82 and 5.08 ± 0.89 Mg C ha^{-1} y^{-1} in the CF and CM plot, respectively (Table 2). The ANCOVA revealed that the annual F_{H} was not affected by the treatment significantly ($p = 0.402$) (Table 3). Among the tested meteorological factors, only air temperature had a significant positive effect on the annual F_{H} ($p < 0.01$). In all cases, the

annual F_{H} in renovation years (Y07–08 and Y12–13) tended to be higher than that in the years before the renovation (Y06–07 and Y11–12). In renovation years, mainly Y12–13, the F_{H} of the first crop was relatively high compared to non-renovation years (Supplementary material 3).

The seasonal changes in vegetation height show that grass started growing in early spring (mid-March) and stopped growing in late autumn (around mid-November) (Fig. 2). The grass growing season coincided closely with the duration of air temperature $> 5.0^{\circ}\text{C}$. In most cases, the grass grew the tallest during the second crop, up to 1.2 m, followed by the first or third crop. During the fourth crop, vegetation height was at most 0.4 m, except in Y08–09 which was up to 0.5–0.6 m.

Annual F_{FYM} varied depending on the year (range: 2.06–6.39 Mg C ha^{-1} y^{-1} ; mean $\pm$ SD: 4.04 ± 1.01 Mg C ha^{-1} y^{-1} ; Table 2).

3.3 Seasonal variations in CO_2 exchange

The annual remaining rate of the NEE data after applying

Table 3. ANCOVA results for the of annual net ecosystem carbon balance components during the study period.

	Factor	Estimate	SE	DF	SS	MS	<i>F</i> -value	<i>p</i> -value	
GPP	Intercept	−97.250	27.570	1	19.216	19.216	12.445	0.006	**
	Plot	1.091	0.664	1	4.162	4.162	2.696	0.135	
	<i>Ta</i>	9.132	2.237	1	25.734	25.734	16.666	0.003	**
	<i>Pr</i>	0.0017	0.0021	1	1.121	1.121	0.726	0.416	
	PPFD	0.0006	0.0005	1	1.958	1.958	1.268	0.289	
	Residuals			9	13.897	1.544			
RE	Intercept	−84.510	25.250	1	14.512	14.512	11.206	0.009	**
	Plot	1.823	0.608	1	11.637	11.637	8.986	0.015	*
	<i>Ta</i>	7.500	2.049	1	17.359	17.359	13.414	0.005	**
	<i>Pr</i>	0.0014	0.0019	1	0.729	0.729	0.563	0.172	
	PPFD	0.0010	0.0005	1	5.292	5.292	4.086	0.074	
	Residuals			9	11.656	1.295			
NEP	Intercept	−37.820	13.140	1	1.673	1.673	8.280	0.021	*
	Plot	49.430	17.130	1	1.684	1.684	8.332	0.020	*
	<i>Ta</i>	3.702	1.074	1	2.399	2.399	11.874	0.009	**
	<i>Pr</i>	0.0003	0.0007	1	0.042	0.042	0.207	0.661	
	PPFD	−0.0004	0.0002	1	0.812	0.812	4.021	0.080	
	Plot * <i>Ta</i>	−4.140	1.413	1	1.734	1.734	8.582	0.019	*
	Residuals			8	1.616	0.202			
F_{H}	Intercept	−40.060	13.170	1	3.261	3.261	9.248	0.014	*
	Plot	0.279	0.317	1	0.273	0.273	0.773	0.402	
	<i>Ta</i>	3.728	1.069	1	4.289	4.289	12.165	0.007	**
	<i>Pr</i>	0.0014	0.0010	1	0.680	0.680	1.928	0.198	
	PPFD	−0.0003	0.0002	1	0.365	0.365	1.036	0.335	
	Residuals			9	3.173	0.353			
NECB	Intercept	14.708	18.954	1	0.440	0.440	0.602	0.458	
	Plot	2.797	0.457	1	27.375	27.375	37.500	< 0.001	**
	<i>Ta</i>	−1.109	1.538	1	0.379	0.379	0.520	0.489	
	<i>Pr</i>	0.0002	0.0014	1	0.010	0.010	0.014	0.909	
	PPFD	−0.0003	0.0004	1	0.373	0.373	0.511	0.493	
	Residuals			9	6.570	0.730			

SE, standard error; DF, degrees of freedom; SS, sum of squares; MS, mean squares; * $p < 0.05$; ** $p < 0.01$; GPP, gross primary production; RE, ecosystem respiration; NEP, net ecosystem production; F_{H} , the amount of carbon removed through harvest; NECB, net ecosystem carbon balance; *Ta*, air temperature; *Pr*, precipitation. The seven-year data were analyzed, except in the years just before grassland renovation (Y06–07 and Y11–12) and the years of grassland renovation (Y07–08 and Y12–13), when annual durations differed substantially from those of the other years. CF and CM plots were coded as dummy variables 0 and 1, respectively.

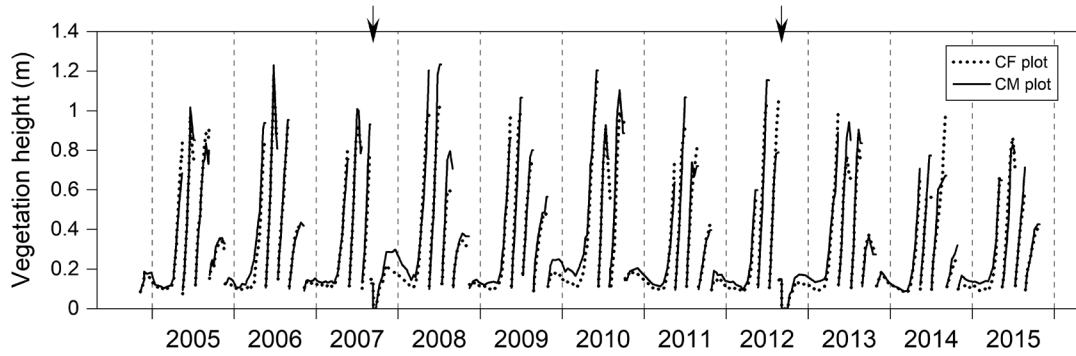


Fig. 2. Seasonal variations in vegetation height. Arrows indicate the dates of plowing during grassland renovation work.

the quality control was 47.3%–65.5% (60.1% for the whole study period) and 58.7%–68.7% (63.6%) in the CF and CM plot, respectively. For the predominant wind direction, the mean distances from the eddy covariance systems at which the cumulative crosswind integrated footprint was equal to 0.8 were 62.9 and 61.9 m in the CF and CM plot, respectively. Thus, >80% of the observed fluxes originated from within the minimal fetch of the experimental plots.

By applying a flux partitioning and gap-filling procedure, daily C fluxes for the entire study period were obtained (Fig. 1). Daily GPP values increased with grass growth and decreased after cutting (Fig. 1 and 2). Daily GPP remained positive even during autumn–winter, although the GPP values were small, indicating that the vegetation was photosynthetically active. Daily RE also changed depending on grass biomass, but this change was driven, to a great extent, by the air temperature. Consequently, daily NEE showed an increase in CO_2 uptake with grass growth and temporary CO_2 release after grass cutting. After grassland renovation was conducted in Y07–08 and Y12–13, significant CO_2 release was observed.

Relationships between seasonal GPP and RE were plotted on a scatter diagram to illustrate the characteristics of seasonal C exchange (Fig. 3; Supplementary material 3). In Fig. 3, the data points below the 1:1 diagonal ($\text{GPP} > \text{RE}$) indicate net C uptake from the atmosphere to the grassland ($\text{NEP} > 0$), whereas those above the diagonal ($\text{GPP} < \text{RE}$) indicate net C release from the grassland to the atmosphere ($\text{NEP} < 0$). For the first crop, data were specifically distributed below the diagonal; the NEP values of this season were positive in all the years in both plots, averaging 2.32 ± 0.73 and $2.53 \pm 0.53 \text{ Mg C ha}^{-1} \text{ period}^{-1}$ in the CF and CM plot, respectively. Furthermore, the data point distribution overlapped to a large extent between the CF and CM plots. During autumn–winter and periods from the second to fourth crop, the data points were positioned close to the 1:1 diagonal. The NEP values of autumn–winter were positive/negative for about half of the years in both plots, averaging -0.02 ± 0.31 and $0.27 \pm 0.48 \text{ Mg C ha}^{-1} \text{ period}^{-1}$ in the CF and CM plot, respectively. In most cases, the NEP values were positive in the second and fourth crops; the NEP values in the CF and CM plots were 0.74 ± 0.49 and $0.18 \pm 0.62 \text{ Mg C ha}^{-1} \text{ period}^{-1}$ in the second crop and 0.37 ± 0.35 and $0.31 \pm 0.40 \text{ Mg C ha}^{-1} \text{ period}^{-1}$ in the fourth crop, respectively. In the

third crop, although the NEP in the CF plot was positive in most cases ($0.36 \pm 0.61 \text{ Mg C ha}^{-1} \text{ period}^{-1}$), the NEP in the CM plot was negative in most cases ($-0.36 \pm 0.53 \text{ Mg C ha}^{-1} \text{ period}^{-1}$). For the second and third crop, the data for the CM plot tended to be distributed above that of the CF plot; this tendency was more pronounced for the third crop.

As shown in Fig. 4, a scatter diagram showing seasonal NEP versus F_H clarified seasonal NECB (also see Supplementary material 3). The NECB for autumn–winter was equal to NEP because this season had no F_H and was positive/negative in about half of the years in both plots, as mentioned earlier. Most data for the first crop were distributed below the 1:1 diagonal ($\text{NEP} > F_H$), indicating that the NECB for this season was positive (0.38 ± 0.55 and $0.41 \pm 0.64 \text{ Mg C ha}^{-1} \text{ period}^{-1}$ in the CF and CM plot, respectively; the same trend is described in subsequently used parentheses). Although, the data were scattered widely compared with that in the GPP–RE chart, the data point distribution overlapped between the CF and CM plots. For the second and third crops, all data were distributed above the diagonal ($\text{NEP} < F_H$), and the seasonal NECB was negative (second crop: -0.76 ± 0.36 and $-1.31 \pm 0.48 \text{ Mg C ha}^{-1} \text{ period}^{-1}$, respectively; third crop: -0.78 ± 0.52 and $-1.36 \pm 0.37 \text{ Mg C ha}^{-1} \text{ period}^{-1}$, respectively). The data of the CM plot tended to be distributed above that of the CF plot during these seasons, and this tendency was more pronounced in the third crop, as was the case in the GPP–RE chart. In most cases, the NECB of the fourth crop was negative in both plots (-0.19 ± 0.36 and $-0.35 \pm 0.39 \text{ Mg C ha}^{-1} \text{ period}^{-1}$, respectively).

3.4 Annual NECB and its components

The annual NEP, GPP, and RE were 3.66 ± 0.77 , 21.8 ± 3.83 , and $18.1 \pm 3.42 \text{ Mg C ha}^{-1} \text{ y}^{-1}$ in the CF plot and 2.84 ± 0.74 , 22.7 ± 3.50 , and $19.8 \pm 3.51 \text{ Mg C ha}^{-1} \text{ y}^{-1}$ in the CM plot, respectively (Table 2). The annual F_H was always higher than the annual NEP in both plots, resulting in negative NECB values in the CF plot ($-1.32 \pm 0.82 \text{ Mg C ha}^{-1} \text{ y}^{-1}$). In the CM plot, where C was input through FYM application, annual NECB was always positive except in Y12–13 ($1.79 \pm 1.59 \text{ Mg C ha}^{-1} \text{ y}^{-1}$). Across the study period at both plots, the annual NECB was lowest in Y12–13 (-3.25 and $-1.38 \text{ Mg C ha}^{-1} \text{ y}^{-1}$ in the CF and CM plot, respectively). The ANCOVA revealed an interplot difference in the annual RE ($p < 0.05$), NEP ($p < 0.05$),

and NECB ($p < 0.01$). However, the treatment had no significant effect on the annual GPP ($p = 0.135$). Regarding meteorological conditions, air temperature alone have significant positive effects on the annual GPP ($p < 0.01$), RE ($p < 0.01$), and NEP ($p < 0.01$) (Table 3). The interaction effect was detected only between the treatment and air temperature on the annual NEP ($p < 0.05$).

3.5 Carbon exchange after grassland renovation

After the cutting of the third crop (September 1, 2007, and August 22, 2012), the cumulative NEP decreased during and after the renovation work until reaching a trough 23–24 and 17 days after seedling emergence in Y07–08 and Y12–13,

respectively (Fig. 5). The lowest cumulative NEP value was smaller in the CM plot than that in the CF plot in both years. At both plots, the cumulative NEP continued to increase during autumn–winter, and it became positive at approximately the same time in all cases, i.e., in late-March or early-April.

Compared with Y07–08, vegetation was more developed after renovation in Y12–13, with more vigorous growth of sown grass, less bare soil, and fewer emergent weeds (Supplementary material 4). Predominantly, the weed species identified after renovation were *Digitaria adscendens* (H.B.K.) Henr. and *Rumex japonicus* Houtt. During autumn–winter, the vegetation height of the sown grass increased to a peak in mid-December (Y07–08 at

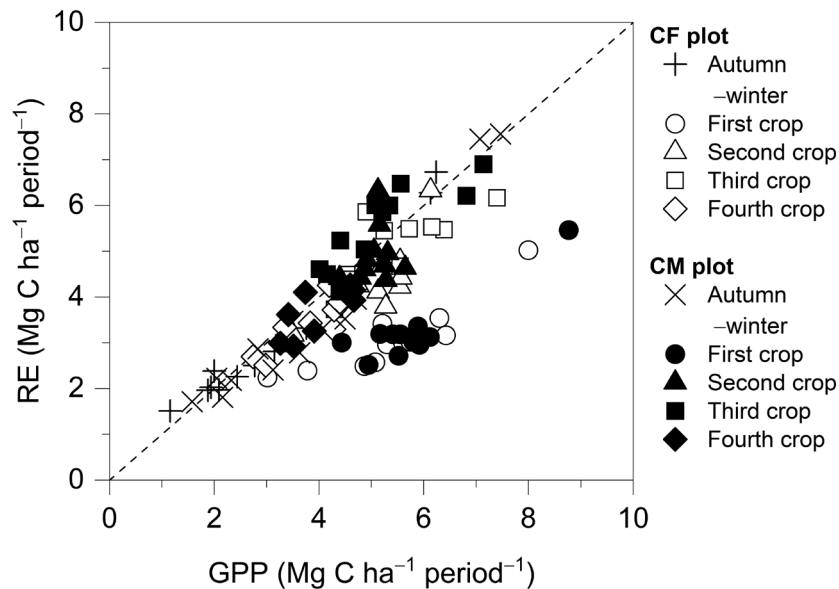


Fig. 3. Relationship between gross primary production (GPP) and ecosystem respiration (RE) for each divided period. The 1:1 diagonal is shown as a dashed line. Data points below the 1:1 diagonal indicate C uptake from the atmosphere to the grassland ($GPP - RE = \text{net ecosystem production} > 0$).

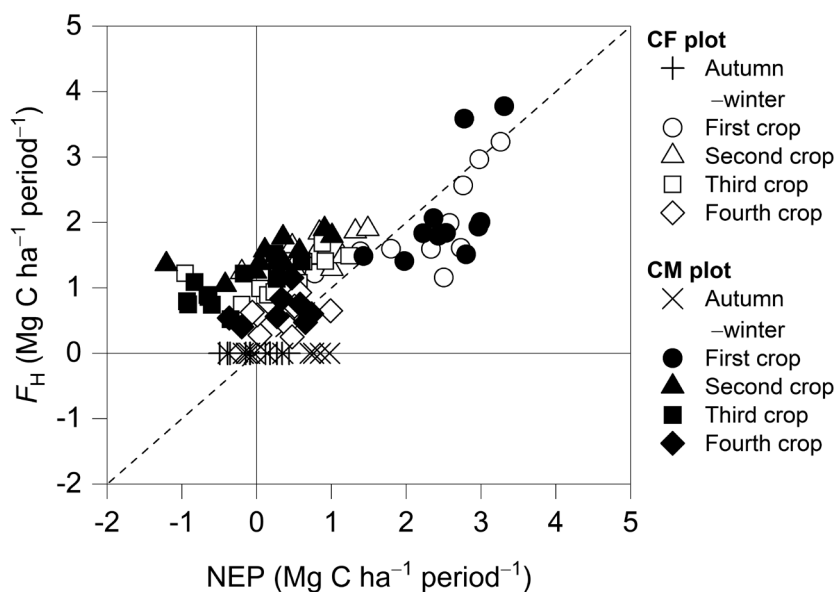


Fig. 4. The relationship between net ecosystem production (NEP) and the amount of carbon removed through harvest (F_H) for each divided period. The 1:1 diagonal is shown as a dashed line. Data points below the 1:1 diagonal indicate that the net ecosystem carbon balance of the period ($NEP - F_H$) was positive.

the CF plot) or mid-November (all other cases) (Supplementary material 5). The vegetation height during autumn–winter was greater in the CM plot than in the CF plot and greater in Y07–08 than in Y12–13.

3.6 SOC concentration and stocks

For all the sampling events, there was no significant interplot difference in the SOC concentration of each layer or SOC stocks of 0–30 cm depth (Table 4). Among sampling events, no significant difference in SOC stocks of 0–30 cm depth was detected. There was also no significant difference in the

SOC concentration of each layer among sampling events with two exceptions: in the CM plot, the SOC concentration of the 0–5 cm layer analyzed just after renovation (Y12–13, November 2012) and in the next year (Y13–14, November 2013) were significantly decreased from just before the renovation (Y11–12, July 2012). However, SOC concentration of the 0–5 cm layer in the CM plot analyzed in Y13–14 (November 2014) was recovered to the level at which no statistically significant difference from before the renovation was detected. The results of the statistical analysis also indicated an increasing tendency of SOC concentration in the CM plot before renovation, from

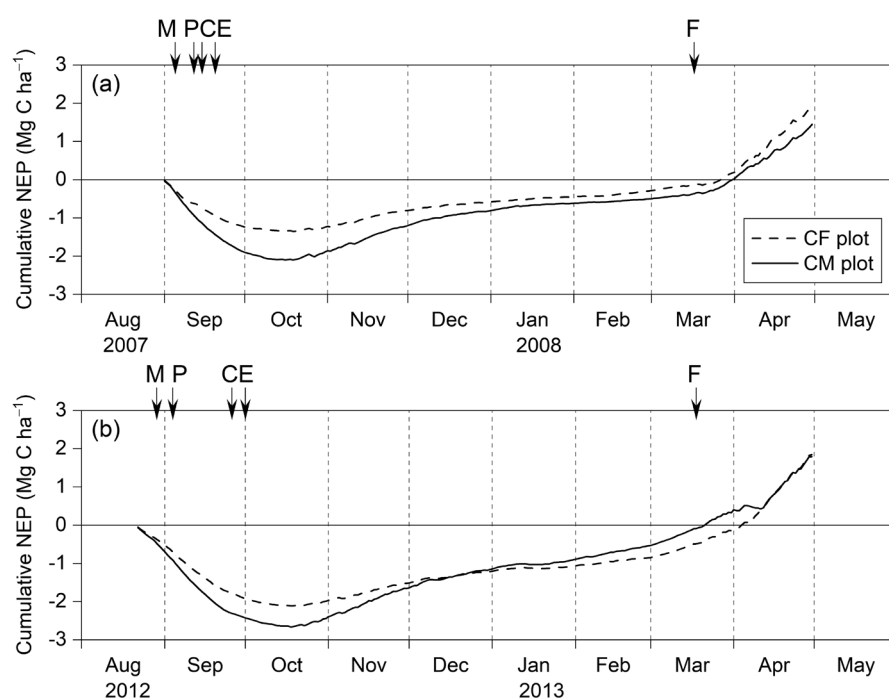


Fig. 5. Changes in cumulative net ecosystem production (NEP) after the cutting of the third crop just before grassland renovation in (a) Y07–08 and (b) Y12–13. M, FYM application; P, plowing; C, completion of grassland renovation; E, seedling emergence; F, chemical fertilizer application in early spring.

Table 4. Changes in soil organic carbon concentration and stocks.

Plot	Year	Sampling date	Carbon concentration (%)				Carbon stocks (Mg C ha ⁻¹)
			0–5 cm	5–10 cm	10–20 cm	20–30 cm	0–30 cm
CF plot	Y10–11	November 11–16, 2010	4.65 ± 3.72	4.25 ± 1.31	4.04 ± 1.11	3.72 ± 0.74	114.3 ± 15.8
	Y11–12	July 23–24, 2012†	5.72 ± 4.45	4.84 ± 1.50	4.99 ± 1.35	4.45 ± 0.96	137.4 ± 21.8
	Y12–13	November 20–21, 2012	4.79 ± 4.46	4.92 ± 1.74	5.26 ± 1.63	4.46 ± 1.16	135.8 ± 30.2
	Y13–14	November 6, 2013	4.67 ± 4.12	4.56 ± 1.43	4.43 ± 1.38	4.12 ± 1.54	123.4 ± 31.7
	Y14–15	November 6, 2014	5.12 ± 3.72	4.58 ± 1.54	4.53 ± 1.18	3.72 ± 0.79	121.0 ± 17.2
CM plot	Y10–11	November 11–16, 2010	5.22 ± 2.89 ^{ab}	3.83 ± 0.64	3.75 ± 0.81	2.89 ± 1.04	110.9 ± 17.5
	Y11–12	July 23–24, 2012†	5.86 ± 2.82 ^a	3.71 ± 1.07	4.03 ± 1.04	2.82 ± 1.37	117.8 ± 31.0
	Y12–13	November 20–21, 2012	3.55 ± 3.19 ^b	3.62 ± 0.83	4.64 ± 1.77	3.19 ± 1.19	114.4 ± 36.7
	Y13–14	November 6, 2013	3.83 ± 2.60 ^b	3.93 ± 0.73	3.89 ± 1.04	2.60 ± 1.54	109.9 ± 24.1
	Y14–15	November 6, 2014	4.45 ± 2.37 ^{ab}	3.89 ± 0.90	4.61 ± 1.56	2.37 ± 1.41	110.6 ± 25.8

Values indicate mean ± standard deviations (n = 5). SOC stocks were quantified by equivalent soil mass in each plot; the soil mass in Y10–11 was chosen as the reference mass for each plot. No significant interplot difference was found in the SOC concentration of each layer or SOC stocks. A significant difference was found between values with different letters ($p < 0.05$) within each plot.

† Just before grassland renovation was conducted in Y12–13.

Y10–11 (November 2010) to Y11–12 (July 2012).

4. Discussion

4.1 Annual and seasonal NECB

The fertilization regime used in this study was designed to ensure that the nutrient supply to each plot was comparable while one plot received FYM application annually. No interplot difference was found in annual F_H (and GPP; Table 3), indicating that the fertilization regime worked well. Concurrently, the different fertilization regimes used in each plot resulted in statistically significant interplot differences in annual RE, i.e., the sum of plant respiration and soil organic matter decomposition. Thus, we can assume that the level of annual plant respiration was the same at both plots because no significant difference in annual grass biomass production existed between the plots. We can also assume that the decomposition of soil organic matter, other than that applied via FYM, was at the same level in each plot. Therefore, the difference in the annual RE can be regarded as the C derived from the decomposition of applied FYM in the CM plot. The interaction effect between the treatment and air temperature was detected on the annual NEP (Table 3). However, it is difficult to interpret the interaction effect because no interaction was detected on its components, GPP and RE. We, thus, focused on GPP and RE rather than NEP when considering the effect of air temperature to avoid the discussion become too complicated.

The annual F_H always exceeded the annual NEP in both plots; therefore, a certain amount of C was lost from the grassland without organic matter application at the site, as shown by the negative NECB in the CF plot. Although the annual RE in the CM plot was higher than that in the CF plot, the annual NECB values were positive, excluding Y12–13 (Table 2). Thus, application of FYM at the recommended level increased grassland C stock at the study site, resulting in significant interplot difference in annual NECB. Although data are limited, some studies have assessed NECB in managed grasslands in relation to cutting. In agreement with our findings, most previous studies found that NECB was negative without organic matter application (e.g., -1.93 to 0.80 Mg C ha⁻¹ y⁻¹), but became positive when organic matter was applied (e.g., 0.80 to 6.83 Mg C ha⁻¹ y⁻¹) (Supplementary material 6; Ammann *et al.*, 2007; Prescher *et al.*, 2010; Hussain *et al.*, 2011; Beetz *et al.*, 2013; Hirata *et al.*, 2013; Limin *et al.*, 2015). The annual NECB estimated in the present study was within the range of the estimates reported in the above-referenced studies.

A scatter diagram of seasonal GPP versus RE during the 11-year study period showed that the magnitude of GPP in the first crop specifically exceeded that of RE (Fig. 3; Supplementary material 3). This can be explained by the high GPP resulting from the vigorous grass growth during this season and the small RE driven by the relatively low air temperature compared with those during the other cropping seasons ($10.1 \pm 0.65^\circ\text{C}$; Supplementary material 2). Substantial overlap of the data points in this season indicated that the CF and CM plots had similar magnitudes of GPP or RE, suggesting that the rate of FYM decomposition was relatively low, likely due to low air temperature. For the second and third crops, the data points of the CM plot were largely distributed above those of the CF plot, which reflected a higher FYM decomposition

rate, likely due to the higher air temperature during these seasons ($19.1 \pm 1.38^\circ\text{C}$ and $23.3 \pm 0.62^\circ\text{C}$ in the second and third crop, respectively; Supplementary material 2). Specifically, in the CM plot, the magnitude of the RE exceeded that of the GPP in the third crop in most years, resulting in negative NEP values on average for 11 years. Although the magnitude of the GPP and RE in the fourth crop was relatively small compared with that of the first to third crops at both plots, the magnitude of NEP during this period was not small and was comparable to that of the third crop. This is likely due to relatively low air temperature during this season ($12.3 \pm 7.94^\circ\text{C}$), causing smaller RE compared to GPP. The annual NEP depends largely on the NEP of the first crop, as indicated by Matsuura *et al.* (2014); the magnitude of first crop NEP occupied 54% and 60% of the magnitude of annual NEP on average in CF and CM plot, respectively.

Given our analysis of 11 years of data, the characteristics of seasonal NECB were represented more clearly in the present study than was possible in the study of Matsuura *et al.* (2014) (Fig. 4; Supplementary material 3). In the current study, the seasonal NECB for the first to third crops was similar to that reported by Matsuura *et al.* (2014), i.e., the seasonal NECB was positive in most cases in the first crop and negative in the second and third crops. However, unlike the earlier study, the NECB of the fourth crop was negative in most cases in the present study, and the NECB in autumn–winter was positive/negative in half of the cases. Both experimental plots had a similar magnitude of NECB during the first crop period; however, during the second and third crop periods, the NECB was lower in the CM plot than in the CF plot, which was likely due to the large RE in the former during summer, as indicated earlier.

Despite the net C loss and gain estimated from the flux measurements in the CF and CM plot, respectively, an increase or decrease in the SOC stocks in these plots was not detected. This discrepancy is probably due to errors both in flux measurements and repeated soil sampling. Uncertainties associated with flux measurements can cause overestimation and underestimation of NEP. The random error of the flux measurements using eddy covariance was estimated to be 10%–30% (Rannik *et al.*, 2016). Rutledge *et al.* (2015) conducted uncertainty evaluation of eddy covariance and estimated the uncertainty of annual NEP, including random and systematic errors, as 23.1%–28.6%, with a value of 104.5% in a year with small annual NEP value of 0.61 Mg C ha⁻¹ y⁻¹. In addition, if the C is lost via leaching, which was not considered in this study, that would lead to an underestimation of NEP. Sparling *et al.* (2016) reported that annual leaching losses of dissolved organic C from managed grassland contributed to 2%–5% of NECB. Furthermore, quantifying SOC changes in <5 years with repeated soil sampling is difficult due to a generally large background SOC pool and high spatial variability (Smith, 2004; Rees *et al.*, 2005). Moreover, data variability in the SOC stocks becomes larger in stony and grassland soils than in non-stony and cropland soils (Goidts *et al.*, 2009). The cumulative NECB from Y10–11 to Y13–14, which is comparable to the duration of the soil sampling events from Y10–11 (November 2011) to Y14–15 (November 2014), was -7.98 and 3.88 Mg C ha⁻¹ in the CF and CM plot, respectively (Table 2). The SD of the soil sampling data was 15.8 – 31.7 and 17.5 – 36.7 Mg C ha⁻¹ in the CF

and CM plot, respectively (Table 4); the errors were likely too large to detect changes in the SOC stock during this period. In the CM plot, undecomposed coarse fragments of annually applied FYM, which were accumulated on the mineral soil surface, were not collected as soil samples; therefore, the C contained in them was not evaluated in SOC measurements. This may have led to the data discrepancy in the CM plot. In addition, if leached dissolved organic C accumulated in subsoil >30 cm, this can cause an underestimation of SOC stocks.

However, although not statistically significant, the increasing tendency of the SOC concentration in the 0–5 cm layer of the CM plot (Table 4) suggests that FYM application can increase SOC. The results of the annual NECB obtained in this study agree with the report on SOC monitoring in nine managed grassland sites across Japan (Matsuura *et al.*, 2021). They show that FYM application at the rate of 30 Mg ha⁻¹ y⁻¹ increases SOC stocks of 0–30 cm depth at all nine sites, while SOC stocks decrease without FYM application at four sites, based on five-year SOC monitoring and statistical model analysis.

4.2 Effects of annual FYM application

Given the statistically significant interplot difference in annual RE (Table 3), we herein define the difference in RE between the two plots as follows:

$$\Delta RE = RE_{CM} - RE_{CF} \quad (3)$$

where _{CF} and _{CM} indicate the plot names. Given that both the annual plant respiration rate and decomposition of soil organic matter, other than that applied via FYM, for the two plots were at the same level, we regard annual ΔRE as the amount of decomposed organic C in F_{FYM} . If the applied FYM decomposed completely, the $\Delta RE / F_{FYM}$ ratio would theoretically converge to 1.0 over time. However, the ratio almost reached a plateau around Y09–10 or Y10–11, i.e., 6–7 years after annual FYM

application treatment was started, although large variation existed in annual $\Delta RE / F_{FYM}$ (Fig. 6). From Y09–10 to Y14–15, the mean annual $\Delta RE / F_{FYM}$ was 0.55. This suggests that 45% of organic C in annually applied FYM remained in the soil after reaching the $\Delta RE / F_{FYM}$ plateau.

Oyanagi *et al.* (2011) evaluated the recalcitrant fraction of cattle FYM using acid detergent lignin as an indicator; they reported its content as 20%–30% of the mass of FYM. Fujiwara *et al.* (2009) reported the acid detergent lignin content of cattle FYM as 31.6%–32.6%. Thus, our estimate of the remained fraction of applied FYM in the soil was slightly higher than previous estimates. Organic matter can bind with reactive metal phases in the soil to form organo-metal complexes (Boudot *et al.*, 1989; Wagai *et al.*, 2020) that contribute to SOC stabilization (Torn *et al.*, 1997). It is possible that, in addition to the accumulation of the recalcitrant fraction of applied FYM, stabilization of C from FYM occurred in the soil at our study site. Hopkins *et al.* (2009) reported a SOC increase of 2–8 Mg C ha⁻¹ y⁻¹ in UK grasslands subjected to long-term FYM application, implying that some part of the FYM applied to the soil surface was stabilized by soil minerals. Several studies have found that long-term FYM application in cropland promotes SOC stabilization through interactions with soil minerals (Ludwig *et al.*, 2010; Bhattacharyya *et al.*, 2011; Ghosh *et al.*, 2018). The stabilization of C from applied FYM was found to be higher in the subsoil layer than in the topsoil (Bhattacharyya *et al.*, 2011), and some level of soil mixing or tillage may be effective for SOC stabilization (Dal Ferro *et al.*, 2020), as discussed in the next subsection.

From Equations 1 and 2, the NECB of each plot is expressed as follows:

$$NECB_{CM} = GPP_{CM} - RE_{CM} - F_{H,CM} + F_{FYM,CM} \quad (4)$$

$$NECB_{CF} = GPP_{CF} - RE_{CF} - F_{H,CF} \quad (5)$$

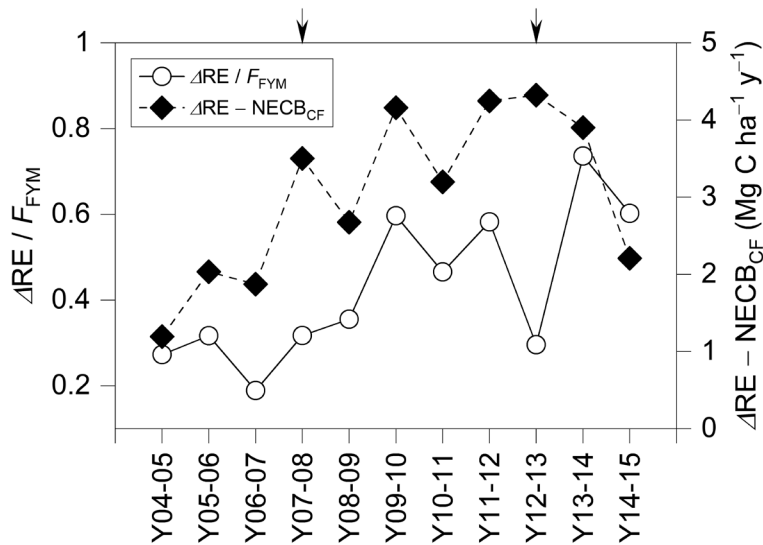


Fig. 6. Changes in the ratio of decomposed C in the amount of C input through farmyard manure (FYM) application (F_{FYM}), and the minimum F_{FYM} required to make net ecosystem carbon balance (NECB) in the CM plot ($NECB_{CM}$) positive. ΔRE is the difference in ecosystem respiration (RE) between the two plots ($RE_{CM} - RE_{CF}$, where the _{CF} and _{CM} indicate the plot names; Equation 3). The minimum F_{FYM} required to make $NECB_{CM}$ positive is expressed as $\Delta RE - NECB_{CF}$ (Equations 3 to 7). Arrows indicate the years of grassland renovation.

Assuming that $GPP_{CM} = GPP_{CF}$ and $F_{H,CM} = F_{H,CF}$, as discussed above, $NECB_{CM}$ can be expressed using Equations 3, 4, and 5, along with the results from both the plots, as follows:

$$\begin{aligned} NECB_{CM} &= GPP_{CF} - (RE_{CF} + \Delta RE) - F_{H,CF} + F_{FYM,CM} \\ &= NECB_{CF} - \Delta RE + F_{FYM,CM} \end{aligned} \quad (6)$$

The amount of annual $F_{FYM,CM}$ required to make the $NECB_{CM}$ positive will be obtained if the right side of Equation 6 becomes positive, and the condition will be as follows:

$$F_{FYM,CM} > \Delta RE - NECB_{CF} \quad (7)$$

Thus, $NECB_{CM}$ will be positive if an amount of C greater than $\Delta RE - NECB_{CF}$ is annually input via FYM application. The difference in NECB between the plots arises from $F_{FYM,CM}$ and ΔRE (i.e., the amount of decomposed organic C in $F_{FYM,CM}$), as indicated in Equation 6. Accordingly, as shown in Equation 7, positive $NECB_{CM}$ will be achieved with a C input via FYM application corresponding to ΔRE in addition to the amount of C required to make the $NECB_{CF}$ (negative values) zero. In the present study, like the annual ΔRE , annual $\Delta RE - NECB_{CF}$ reached a plateau around Y09–10 (Fig. 6). The mean annual $\Delta RE - NECB_{CF}$ from Y09–10 to Y14–15 (including Y14–15, when the annual $\Delta RE - NECB_{CF}$ suddenly dropped) was $3.67 \text{ Mg C ha}^{-1} \text{ y}^{-1}$; this indicates the minimum $F_{FYM,CM}$ required to make the site a net C sink. Hirata *et al.* (2013) reported that $>2.5 \text{ Mg C ha}^{-1} \text{ y}^{-1}$ of FYM application was required to make NECB positive at four grassland sites in Japan, including the present study site; thus, our estimated value is higher than that estimated by Hirata *et al.* (2013). However, their value was estimated using three years of data following the start of FYM application treatment, and it is consistent with values calculated using the initial three or four years of data collected during the current research (Fig. 6).

4.3 Effects of grassland renovation

In renovation years, annual F_H tended to be higher than that in the years before the renovation (Table 2). However, the effect of renovation on F_H was not subsequently clear. This may be due to annual variations in F_H masking the effect of renovation on harvest in subsequent years. Some studies have found that grass yield tends to increase after renovation (Velthof *et al.*, 2010; Elsaesser, 2012). However, the positive effect on yield likely lasts only a few years (Hatch and Hopkins, 2007; Kayser *et al.*, 2018).

A decrease in the cumulative NEP during and after renovation work (Fig. 5) showed that the net CO_2 release was due to the sward/soil disturbance by the renovation. This net CO_2 release is probably related to the reduction or absence of photosynthetic CO_2 uptake before seedling emergence; CO_2 release also increased likely due to the accelerated soil organic matter decomposition resulting from increased aeration and breakdown of soil aggregates caused by plowing and subsequent renovation work (Grandy and Robertson, 2006; Willems *et al.*, 2011; Rutledge *et al.*, 2014). An increase in the cumulative NEP after a trough was reached reflected an increase in CO_2 uptake that occurred with the growth of sown grass.

The magnitudes of the decrease and increase in cumulative NEP were larger in Y12–13 than those in Y07–08, probably because, in Y12–13, the degree of sward/soil disturbance was

higher (Reicosky, 1997; Kayser *et al.*, 2018), the renovation work lasted longer, and the vegetation was more developed after renovation (Supplementary material 4), which is also closely related to the degree of sward/soil disturbance (Kayser *et al.*, 2018). The greater vegetation height in Y07–08 compared with that in Y12–13 during autumn–winter (Supplementary material 5) was likely due to the emergence of more weeds, which grew faster than the sown grass. When comparing the cumulative NEP between the plots, the magnitudes of the decrease and increase in cumulative NEP were larger in the CM plot than in the CF plot. CO_2 released by the decomposition of annually applied FYM, including that applied during renovation work, can account for the lower trough in the cumulative NEP data for the CM plot relative to that for the CF plot. In addition, the increased CO_2 uptake during the subsequent autumn–winter period in the CM plot can be explained by faster grass growth due to the increased nutrient supply from FYM, which was applied during the renovation in addition to the chemical fertilizer applied at the same rate in both the plots.

The amount of C released during and after the renovation work appeared to be approximately compensated for by the amount of C absorbed by the newly-established sward during autumn–winter (Fig. 5). The level of the annual $NEP - F_H$ in Y07–08 was the same as that in other non-renovation years, and there seemed to be little renovation effect on the annual NECB (Table 2). This may be because the duration of renovation work was short (4 days) and the degree of soil disturbance was low; thus, the proportion of soil organic matter decomposition in RE was similar to that in non-renovation years. In contrast, the annual $NEP - F_H$ or NECB in Y12–13 was more negative than that in the other years; the annual NECB values were negative even in the CM plot. One contributing factor may have been the higher first crop F_H value in Y12–13 compared with the values in other years, resulting in negative NECB in the first crop (Supplementary material 3). Part of this first crop F_H included the C assimilated by well-developed sown grass during autumn–winter following renovation, which compensated for the C released during and after renovation work. In addition, the proportion of soil organic matter decomposition in RE could be larger in Y12–13 than in other years, not only during autumn–winter, but also, to some extent, during the following crop seasons due to the high degree of soil disturbance that occurred during renovation. This resulted in relatively small NEP values compared with F_H values and, in turn, more negative $NEP - F_H$ or NECB. However, the annual $NEP - F_H$ or NECB in the subsequent years (Y13–14 and Y14–15) were at the same level as that measured before the renovation year (from Y08–09 to Y10–11). Among measured meteorological conditions, air temperature significantly affected annual GPP, RE, F_H , and NEP with the interaction effect between the treatment and air temperature (Table 3; note that the years just before renovation and the renovation years were excluded from the analysis for Table 3). The annual air temperature in Y12–13 was 0.4°C higher compared to Y07–08 (Supplementary material 2), which may affect the C dynamics between these two years. However, meteorological conditions did not significantly affect NECB (Table 3). Thus, it can be said that the difference in net C balance between the two years was mainly due to the degree of sward/soil disturbance.

Several studies have been conducted on the immediate effects (hours to days) of grassland renovation on C dynamics, reporting that CO₂ emissions increase immediately after soil disturbance and decrease only after hours (Reicosky *et al.*, 1997; Willems *et al.*, 2011; Matsuura *et al.*, 2017). The increased emissions are likely due to the physical release of CO₂ entrapped in soil pores or dissolved in soil water (Reicosky and Lindstrom, 1995; Reicosky *et al.*, 1997). In contrast, little research has been conducted on the longer-term effect (months to years) of grassland renovation on C dynamics. This is partly because of the inherent challenge in evaluating the effects of renovation on NEP or NECB after vegetation starts to grow post-renovation, i.e., it is difficult to quantify the contribution of plant respiration and soil organic matter decomposition on total RE (Trumbore, 2006). Using eddy covariance, Rutledge *et al.* (2015) found that the annual NEP in grazing pasture remained positive even during a renovation year. Reinsch *et al.* (2018) investigated the effect of grassland renovation on CO₂ emissions using soil respiration measurements and a simulation model; they estimated that a renovation-induced increase in organic matter decomposition was limited to the first 6 months after soil disturbance. Using eddy covariance measurements, Ammann *et al.* (2013) found that the NECB of manure-applied grassland converted from positive to slightly negative for three years following grassland renovation; however, a long fallow period (5 months) during the renovation work may have contributed to this sustained negative effect on NECB. Although the available data were limited, our results were consistent with the findings of these studies; it appeared that the negative effect of grassland renovation on annual NECB may last for a few years at most following renovation (Table 2).

Grassland renovation with full inversion tillage has the potential to increase SOC stock by burying C-rich topsoil into the subsoil region and bringing low-C subsoil to the soil surface, where large amounts of organic matter are supplied from sward production and FYM application (Mukumbuta *et al.*, 2019; Mukumbuta and Hatano, 2020; Lawrence-Smith *et al.*, 2021; Madigan *et al.*, 2022). Our results showed a decrease in SOC concentration at the 0–5 cm layer of the CM plot after renovation (Table 4), indicating that the C-rich 0–5 cm layer of soil was buried into a deeper horizon. Incorporating plant residue and/or applied FYM into the soil can increase the probability of these interacting with soil minerals and contributing to C stabilization, as noted in the previous subsection. Recent studies in New Zealand have provided evidence that SOC increases after grassland renovation with full inversion tillage. The increase rates at the 0–15 cm layer have been estimated at 0.36–0.64 Mg C ha⁻¹ y⁻¹ (inversion depth of 30 cm) (Lawrence-Smith *et al.*, 2021) and 1.2–1.8 Mg C ha⁻¹ y⁻¹ (inversion depth of 170 cm) (Schiedung *et al.*, 2019) in the 20 years following renovation.

5. Concluding remarks

Our analyses, based on 11 years of CO₂ flux measurements, revealed that the annual NECB of the studied grassland site was negative (-1.32 ± 0.82 Mg C ha⁻¹ y⁻¹) without FYM application, after considering C removal through grass harvest. However, the annual NECB became positive (1.79 ± 1.59 Mg C ha⁻¹ y⁻¹)

when the recommended level of FYM was applied annually, although part of the applied FYM was decomposed, which released CO₂ into the atmosphere. In addition, the FYM application did not affect annual F_H by adjusting the application rates of chemical fertilizers. Thus, an appropriate FYM application is recommended as a management practice in temperate managed grasslands that can also contribute to net C gain; indeed, it represents an environmentally sound and sustainable livestock husbandry practice and the effective use of livestock excreta that helps maintain grass production. Our flux measurements also revealed the seasonality of C exchange and net C balance, indicating that the FYM decomposition rate was low during the first crop period when the air temperature was also low but high during the second and third crop periods, i.e., in summer. As the temperature is expected to gradually rise over time, the decomposition of applied FYM could accelerate not only during the second and third crop periods but also during the first crop period. This would weaken the positive effect of the FYM application on the annual NECB, as Matsuura *et al.* (2021) estimated through an analysis of SOC monitoring data from grassland sites across Japan. Grassland renovation with full inversion tillage can affect annual NECB as well as seasonal C exchange when the degree of sward/soil disturbance is high. The effect on annual NECB at our study site was likely limited to renovation years. However, we only collected data on two renovation events. Hence, it is difficult to accurately evaluate the effects of renovation on NEP and NECB during a period of months or years following renovation.

Additional qualitative and quantitative studies evaluating SOC should be conducted to estimate the degree of C stabilization and residence time of C from applied FYM. A continuous and repeated study of grassland renovation should also be performed to further understand the effect of renovation on long-term NECB or SOC stock changes. To resolve the discrepancy between the annual NECB estimated from flux measurements and SOC stock changes based on repeated soil samplings, more studies to evaluate the amount of C loss via leaching, longer-term SOC monitoring, deeper layer SOC stocks, and the amount of C in accumulated FYM on the soil surface are necessary. Although further research described above is required, the present study has improved our understanding of C dynamics in managed grasslands as well as the effects of grassland management practices on NECB.

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